Deepak Mallampati

APPLIED AI ENGINEER

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Kent, OH | US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years training, evaluating, and deploying LLM agents in production at Vanguard (25+ AI agents, 100,000+ daily requests, 27% task-success lift) and Progress Solutions (conductor-subagent orchestration, 42% token reduction, 60% reliability gain). Hands-on with multi-agent systems, post-training (QLoRA on Qwen3-27B / NVIDIA H100), and structured LLM evaluation (RAGAS, BERTScore, custom domain metrics). MS in CS, Kent State (2025). Targeting Scale AI's Agents team to train SOTA enterprise GenAI models and integrate cutting-edge post-training algorithms into production training infrastructure.

CORE COMPETENCIES

Agent Training (Multi-Agent) • Post-Training (QLoRA, Instruction-Tuning) • LLM Evaluation (RAGAS, BERTScore) • RAG + Vector Search • Production LLM Deployment • LangChain / LangGraph • Python, PyTorch, Transformers • AWS, GCP, Azure ML

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Prompt Engineering | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Content Safety | Hugging Face Transformers
Machine Learning & NLP: PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Model Optimization | A/B Testing | Drift Detection | RAGAS Evaluation | BERTScore
Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking Strategies | Semantic Caching
MLOps & Cloud Infrastructure: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services, Content Safety), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model Monitoring & Observability
Data Engineering: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy | Data Lineage & Governance
Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging | Zero-Trust Architecture
Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira, Confluence | Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present*

AI Engineer - USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Designed and operationalized **privacy-preserving data workflows** for sensitive clinical datasets, implementing K-anonymity, L-diversity, and differential privacy (Laplace mechanism) to enable compliant analytics and model development with zero PII exposure.
- Built fault-tolerant **automation infrastructure** featuring scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) paired with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard *Jan 2024 - Jan 2025*

AI Engineer Intern - USA

- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini** models, improving task success rates by **27%** on internal evaluation datasets and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University *Oct 2023 - Jan 2024*

ML & Gen AI Research Assistant - USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset, applying k-anonymity (k=3), l-diversity (l=2), and Laplace differential privacy across four ε levels (0.1-2.0) over 8 clinical features.
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** and eliminated unique-record exposure (0.94% to 0%) under k-anonymity + l-diversity, while retaining **99% of baseline predictive performance** (Random Forest AUC 0.689 vs. 0.696; weighted-F1 0.895).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated 361-example instruction dataset, conditioning the model on a six-stage narrative schema for controllable long-form generation.
- Authored a peer-review-ready **IEEE-format conference paper** documenting the methodology, experiments, and results (LaTeX).

Emerson *Jun 2022 - Apr 2023*

ML Trainee - India

- Developed and trained supervised **regression and classification models** on operational pipeline-storage data, applying standard ML workflows to equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** models for entity extraction and text classification on domain-specific financial datasets, achieving **89% F1-score** on custom NER tasks.

SELECTED PROJECTS

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight - compressing hours of repetitive job-hunting into an automated workflow.

Privacy-Preserving Clinical ML Pipeline

A framework that lets researchers analyze sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy - quantifies exactly how much privacy each technique buys for how little accuracy cost.

Story Director LLM

A fine-tuned language model that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

EDUCATION

M.S. in Computer Science - *Kent State University, OH* *2025*

GPA: 3.70 / 4.0